

Richard Oliveira

curriculum vitæ

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My background can be broadly defined as that of a Mathematical Engineer, with a strong interest in developing data-driven mathematical models to describe complex systems in the fields of Machine Learning, Control Theory and Optimization.

Education

Doctor of Philosophy (Ph.D.) in Data Analytics and Decision Sciences

Politecnico di Milano, Milano, Italy

September 2025 - Present

Master of Science (MSc.) in Electrical Engineering, Signal and Image Processing

Thesis: “On the General Problem of the Pearson-Rayleigh Random Walk”

University of California, San Diego, La Jolla, CA, USA

September 2022 - June 2024

Bachelor of Science (BSc.) in Electrical Engineering

University of California, San Diego, La Jolla, CA, USA

September 2020 - June 2022

Research Experience

Graduate Student Researcher

Politecnico di Milano, Electronics, Information Technology, and Bioengineering Dept.

Advisor: Dr. Stefano Savazzi, Prof. Monica Nicoli

September 2025 - Present

- ❖ Mathematical foundations of machine learning, focusing on consensus algorithms and implicit regularization.

Graduate Student Researcher

University of California, San Diego, Electrical and Computer Engineering Dept.

Advisor: Prof. Massimo Franceschetti

September 2023 - August 2024

- ❖ Contributed to the theory of Discrete Pearson-Rayleigh Random Walks by analyzing how walks of this kind behave in Discrete Space with Geometric and Poisson distributed step sizes.

Undergraduate Student Researcher

University of California, San Diego, Electrical and Computer Engineering Dept.

Advisor: Prof. Nick Antipa

September 2020 - June 2022

- ❖ Analysis and development of a Linear, Spatial - Invariant Imaging system (Diffuser Cam).
- ❖ Experimentally recovered a Point Spread Function of the Imaging system.
- ❖ Implemented in Python an accelerated version of Gradient Descent (ADMM), which was used for the minimization of a Loss in the context of solving an Inverse Problem (Scene Reconstruction).

Publications

Journal

- R. Oliveira, S. Savazzi. "On Solutions to Discrete-Time Inverse Problems via Sparse Superpositions of Decaying Shifted Heaviside Functions: A Hardy Space Approach." *IEEE Signal Processing Letters*, February 2026.
- R. Oliveira, S. Savazzi. "Distributed Composite Regularization via Dual-Layer Consensus Optimization." *IEEE Transactions on Signal and Information Processing over Networks*, April 2025.

Manuscripts Under Review

- R. Oliveira, S. Savazzi. "On the Stability of Consensus Gradient Dynamics for Regularized Distributed Deep Matrix Factorization: Explicit Bounds." *IEEE Signal Processing Letters*, February 2026.

Projects

Inverse Problems in NeuroMANCER, November 2024 - December 2024

The NeuroMANCER library enables the incorporation of physics-informed constraints into convex optimization problems. I utilized this library to solve an inverse problem using the MNIST dataset, specifically to evaluate its performance with non-smooth regularizers. When comparing the results with established libraries like CVXPY, I discovered that NeuroMANCER requires careful fine-tuning to effectively handle non-differentiable regularizers.

Deep Pose Estimation from 3D Point Clouds, March 2024 - June 2024

In this work, we develop a NN to predict relative poses between two sets of 3D point clouds from LiDAR scans and RGBD camera data, obtained from the EDEN Data Set. It aims to improve upon traditional point cloud registration methods, such as the Iterative Closest Point (ICP) algorithm by addressing two main challenges: sensitivity to initialization and the high computational cost of successive data alignment and optimization steps.

Visual-Inertial SLAM, January 2024 - March 2024

In this project, we present a way of solving the SLAM problem by using the Extended Kalman Filter (EKF). Initially, we simply use the *Prediction Step* of the EKF to track the position of the robot over time (Localization). Then, in order to estimate the location of landmark positions measured by a stereo camera mounted on the robot, we only use the *Update Step* of the EKF (Mapping). Lastly, we combine the *Prediction* and *Update Step* of the EKF such that we perform Localization and Mapping simultaneously, hence we perform Visual-Inertial SLAM.

Li-DAR-based SLAM, January 2024 - March 2024

In this project we present a way of approaching the SLAM problem for a Differential-Drive robot via the use of measurements coming from wheel encoders, IMU, LiDAR and a RGBD camera (Kinect). We make use of the Iterative Closest Point (ICP) algorithm to estimate the robot's pose over time while using LiDAR measurements and also use GTSAM which re-formulates the SLAM problem and optimizes the trajectory of our robot. We then build an occupancy grid of the environment, which addresses the Mapping part of SLAM, using an optimized trajectory obtained from ICP and LiDAR measurements, and GTSAM.

A Literature Review of the Theory and Applications of Convex Lifts, January 2023 - March 2023

This literature review offers a concise overview of the theory and applications related to lifts of convex sets. A lift of a convex set is a higher-dimensional convex set that can be linearly projected onto the original set. In many cases, lifting a convex set to a higher dimension simplifies its structure significantly. Lifts of this kind are particularly important in the field of optimization.

Teaching

ECE 227: Big Network Data - Graduate Teaching Assistant

University of California, San Diego, Electrical and Computer Engineering Dept.

March 2024 - June 2024

- ❖ A course on Graph Analysis, in which I developed course projects in the context of Robotics.
- ❖ Provided feedback and instructional help to students on the course projects during lecture hours.

ECE 35: Introduction to Analog Design - Undergraduate Teaching Assistant
University of California, San Diego, Electrical and Computer Engineering Dept.
September 2021 - December 2021

- ❖ Provided academic support to students during lecture hours.
- ❖ Developed instructional materials and attended lab sessions.
- ❖ Assisted in grading exams and held multiple weekly Office Hours during the quarter.

Employment

Researcher
Italian National Research Council (CNR), Institute of Electronics, Computer and Telecommunication Engineering (IEIIT), Milano
September 2025 - Present

Distributed optimization theory, regularization, consensus networks.

Junior Data Scientist
Onikron LLC, Remote
September 2024 - August 2025

Implemented and researched modern NLP models, such as Transformers and RNNs with PyTorch. Developed Lemmatization and Tokenization scripts for Data Pre-Processing, as well as Clustering for Unsupervised Learning.

Algorithms & Signal Processing Engineer
General Atomics A.S.I, San Diego, CA
June 2023 - August 2023

Research modern NN architectures for the development of a Deep Learning model in the context of Computer Vision. Additionally, I have created a Data Visualization tool in Python, using PythonBokeh, for the analysis of heat maps. This tool enabled the team to perform appropriate Data Pre-Processing.

DataBase Engineer
Ciena, San Jose, CA
May 2021 - August 2021

Extensive use of SQL queries, through the FLYBID database, which organized sensitive records within the company. Furthermore, special focus was placed on the creation of Data Dictionaries which would simplify Data mapping through the use of LucidChart.

Selected Honors & Awards

Marie Skłodowska-Curie Doctoral Fellowship, September 2025-September 2028
Graduate Student Researcher (GSR) Grant, January 2024-March 2024

Skills

Programming Languages: C, C ++, MatLab, SQL, Python, PyTorch, TensorFlow, Assembly, Scikit-learn.

Languages: Italian (Native & Professional Proficiency), English (Native & Professional Proficiency).